DATA ITEM DESCRIPTION

Title: HEMP PROTECTION SUBSYSTEM (HPS) PERFORMANCE IMMUNITY TEST REPORT

Number: DI-EMCS-81854A	**Approval Date:** 20230320
AMSC Number: 10415	**Limitation:** N/A
DTIC Applicable: No	**GIDEP Applicable:** No
Preparing Activity: DS	**Project Number**: EMCS-2023-010

Use/Relationship:

The HEMP Protection Subsystem (HPS) Performance Immunity Test Report applies to MIL-STD-3023A, *High-Altitude Electromagnetic Pulse (HEMP) Protection for Military Aircraft*. DI-EMCS-81854, *Test Report for Mission Critical Subsystem Immunity Test*, is hereby cancelled and superseded by DI-EMCS-81854A. This data item description (DID) contains the format, content, and intended-use information for the data deliverable resulting from the immunity test tasks as described in MIL-STD-3023A, Appendix C, sections C.4.8 and C.4.9.

The HEMP Protection Subsystem (HPS) Performance Immunity Test Report documents the methods of test-execution, data-recording, analysis, inspection, and reporting that were used by the contractor to assess and document the results of the MCS's performance. For each individual MCS that was immunity tested per MIL-STD-461 (or equivalent), an individual test report shall be prepared. The report provides the means for the Government to evaluate and certify the HEMP-protection performance results of the MCS and any associated special protective measures (SPMs), in that the report demonstrates compliance with respect to MIL-STD-461's (or equivalent) pass and fail requirements and MIL-STD-3023A's test requirements.

The report includes discussion of recommended corrective actions, if needed.

If the MCS has already been MIL-STD-461 (or equivalent) tested prior to its installation into the aircraft, then documentation of previous test plans, reports, and test data, should be attached as an appendix.

This DID is related to DI-EMCS-81850A and DI-EMCS-81853A.

This DID supersedes DI-EMCS-81854.

Requirements:

1. Reference documents. The applicable issue of the documents cited herein, including their approval dates and any applicable amendments, notices, and revisions, shall be as specified in the contract. The HEMP Protection Subsystem (HPS) Performance Immunity Test Report's classification shall be in accordance with any relevant facility- or system-specific classification guides and also the Defense Threat Reduction Agency (DTRA) Security Classification Guide for DoD Electromagnetic Pulse (EMP) Programs and Activities.

(DTRA's security classification guide is available by mail request to ATTN: RD-NTSA, Defense Threat Reduction Agency, 8725 John J. Kingman Road, MSC 6201, Fort Belvoir, VA

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

22060-6201.)

2. Format. Contractor format is acceptable for report (comma-separated values (CSV) is preferred data format).

3. Content. The HEMP Protection Subsystem (HPS) Performance Immunity Test Report documents the test-execution, data-recording, analysis, inspection, and reporting procedures that were used by the contractor to perform the testing on MCS in accordance with MIL-STD-461(or equivalent) and MIL-STD-3023A.

3.1 Administrative data. The HEMP Protection Subsystem (HPS) Performance Immunity Test Report shall contain an administrative section covering the following:

a. Contract number, CDRL number, and date

b. Authentication and certification by a qualified representative of the procuring activity for the test report

3.2 Summary information. HEMP Protection Subsystem (HPS) Performance Immunity Test Report shall summarize the following:

3.2.1 Introduction, to include:

a. General objectives of the test

b. A general description of the MCS and associated SPMs that were tested

c. A statement of any assumptions or limitations associated with the testing

d. A summary of the test pass and fail requirements per MIL-STD-461 (or equivalent)

e. A specific statement whether or not the MCS, with any associated SPMs, passed all requirements, and a summary listing of any requirements that were not passed.

3.2.2 Scope. A general description of overall test-requirements matrix that was used to verify the MCS and any SPM's compliance with MIL-STD-3023A's performance requirements, and an associated listing of the test results, shall be provided. Also, a description of the relative roles of tests, inspections, and analyses shall be provided.

The following topics shall be included:

a. Synopsis of the test procedures performed (as applicable), and references to detailed procedures

b. Statement of any assumptions and limitations associated with MCS immunity tests

c. Pass or failure performance results with respect to all requirements

d. Anticipated impact of any failures

e. Recommendations to resolve any failures

f. Lessons learned for the MCS’s hardness and testability

3.2.3 Methods of demonstration. A general description of the procedures that were used to demonstrate the MCS and associated SPMs’ compliance with each test performance requirement as listed in section 3.3 below.

3.2.4 Engineering factors. Any important engineering factors that affected the test procedures, such as: physical-accessibility constraints of the system’s topology, resource limitations, safety and security protocols, and precedent reports.

3.3 Detailed information. For each MCS being tested, a single test report shall be prepared. The test report shall contain the following information:

a. An identification of the MCS that was tested, including photographs, figures, or diagrams; and including serial numbers, version of software used (if any), and calibration status and due date (if applicable)

b. The location of where the MCS was tested, the company or government agency performing the test, and the testing dates

c. Identification of the test-team personnel and of the Government witnesses that were present

d. A description, with diagrams, showing for the MCS all locations of all points of entry (POEs), external cable interconnections, and a description of any installed HEMP or other electromagnetic protection measures, including SPMs.

e. Descriptions of the source simulator and associated data acquisition equipment, to include the manufacturer, model, and serial numbers of each equipment item; detailed calibration procedures for the equipment; and calibration-traceability documentation

f. Descriptions of the test procedures, including system, subsystem, and circuit configuration requirements; test configuration diagrams; test point locations and descriptions; step-by-step test procedures; and the approach to calculating norms-based design margins

g. Descriptions of methods for measuring noise (instrumentation and local or ambient)

h. A description of models, techniques, and tools used for preliminary analyses and predictions, and their specific application to the specific facility or system to be tested

i. A description of the methods that were applied to quantify and evaluate any failures of the MCS and any associated SPMs’ performance with respect to all pass and fail criteria

j. Identification of the pass and fail criteria, and of any deviations from the requirements of MIL-STD-461 (or equivalent) or MIL-STD-3023A that occurred, along with an explanation of why these deviations were necessary

k. A description of the test schedule as it was executed, including a listing of the prioritization of measurements

l. Descriptions of the data-management procedures that were used, to include: data quality control procedures, data acceptability criteria, annotation and preservation of data records, and the methods to compare results to pass and fail criteria

m. Descriptions of the data-security and physical-security procedures that were applied

n. Photographs and diagrams of the actual test set up, with identification of subsystems and hardware and a distance scale to allow pertinent horizontal/vertical/lateral measurements to be extracted

o. A description of the MCS' operating conditions that were set during the testing (e.g., power-on/off, and any other equipment/cables connected to the MCS and data being transferred and processed during testing)

p. A description of the methods that were used to simulate the operational performance of the MCS during the testing if the actual MCS operation during the testing was impractical

q. Descriptions of how the raw and processed test data were recorded, and the electronic data format used to store the data (CSV is the preferred data format)

r. Summary tables of measured quantities, with calculations and comparisons to pass and fail criteria in MIL-STD-461 (or equivalent) based on norm attributes of each oscillographic waveform, and depictions of any associated onsets of MCS upset or damage

s. Detailed presentation of data, as specified in the "data presentation" sections of the test procedure for in MIL-STD-461 (or equivalent)

t. Copies of measured test results, including all raw and processed test oscillographic waveform test data in engineering units, with a description of the electronic data format and guidance for users to locate specific data within this format

u. Findings from investigations into causes of failures, if any; and corrective actions recommended or, if performed, their re-test results

v. Analysis of the test's adequacy, to include a discussion of the uncertainties involved with the test procedure and measured quantities and any issues regarding the reproducibility of test results

w. Identification of any safety risks encountered, and descriptions of procedures that were used to mitigate these risks, including any electromagnetic radiation hazards and electrical shock hazards

x. Descriptions of lessons learned, to include any insights learned during testing that could be applied to improving HPS performance and improving testing methodologies

y. If applicable, proposed updates to the High-Altitude EMP (HEMP) Hardness Maintenance and Hardness Surveillance (HM/HS) Plan (DID DI-EMCS-81855A) and the MIL-STD-3023A Hardness Allocation Report (HAR) (DI-EMCS-81850A)

z. Test conclusions, including a definitive statement of the hardness of the MCS to the conducted environment, based upon the results of the complete battery of tests and the supporting analyses

aa. References to applicable test plans and procedures

End of DI-EMCS-81854A